

15. Write a Pandas program to keep the rows with at least 2 NaN values in a given DataFrame

CODE:



```
YY.py - C:/Users/hp/AppData/Local/Programs/Python/Python313/YY.py (3.13.9)
File Edit Format Run Options Window Help
import pandas as pd

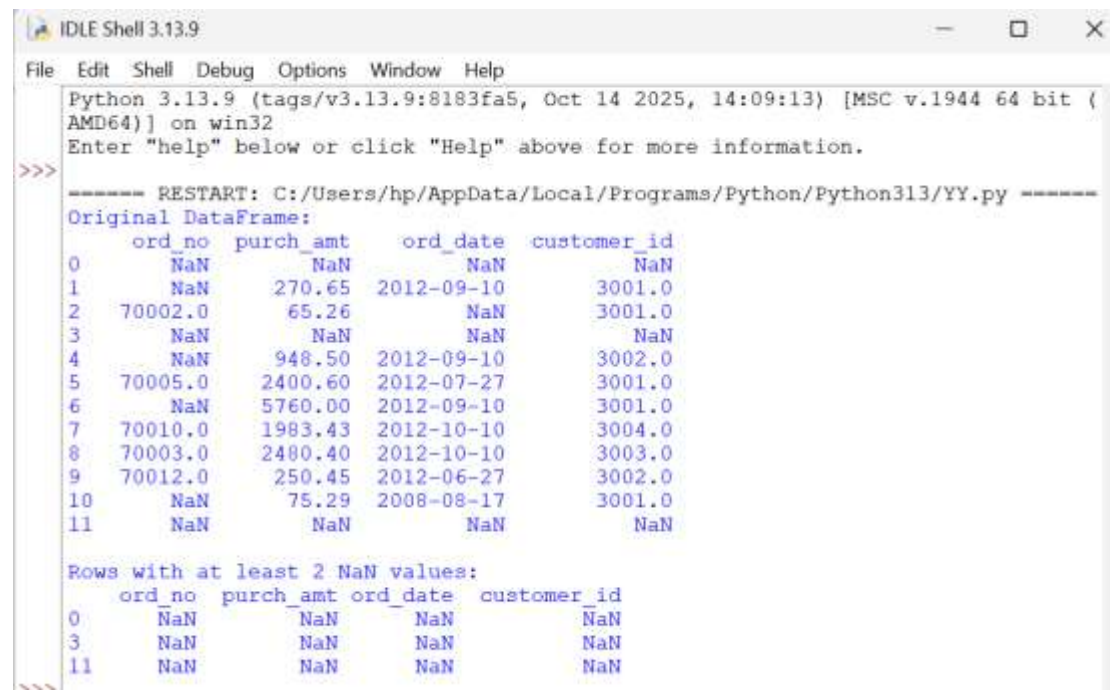
df = pd.read_csv("C:/Users/hp/Downloads/15TH.csv")

print("Original DataFrame:")
print(df)

result = df[df.isna().sum(axis=1) >= 2]

print("\nRows with at least 2 NaN values:")
print(result)
```

OUTPUT:



```
IDLE Shell 3.13.9
File Edit Shell Debug Options Window Help
Python 3.13.9 (tags/v3.13.9:8183fa5, Oct 14 2025, 14:09:13) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> ----- RESTART: C:/Users/hp/AppData/Local/Programs/Python/Python313/YY.py -----
Original DataFrame:
   ord_no  purch_amt  ord_date  customer_id
0      NaN         NaN         NaN         NaN
1      NaN      270.65  2012-09-10       3001.0
2  70002.0       65.26         NaN       3001.0
3      NaN         NaN         NaN         NaN
4      NaN      948.50  2012-09-10       3002.0
5  70005.0      2400.60  2012-07-27       3001.0
6      NaN      5760.00  2012-09-10       3001.0
7  70010.0      1983.43  2012-10-10       3004.0
8  70003.0      2480.40  2012-10-10       3003.0
9  70012.0       250.45  2012-06-27       3002.0
10     NaN        75.29  2008-08-17       3001.0
11     NaN         NaN         NaN         NaN

Rows with at least 2 NaN values:
   ord_no  purch_amt  ord_date  customer_id
0      NaN         NaN         NaN         NaN
3      NaN         NaN         NaN         NaN
11     NaN         NaN         NaN         NaN
>>>
```

16. Write a Pandas program to split the following dataframe into groups based on school code. Also check the type of GroupBy object.

CODE:

A screenshot of a Python IDE window titled "16th.py - C:/Users/hp/AppData/Local/Programs/Python/Python313/16th.py (3.13.9)". The window contains the following Python code:

```
import pandas as pd

df = pd.read_csv("C:/Users/hp/Downloads/16TH.csv")

print("Original DataFrame:")
print(df)

grouped = df.groupby("school")

print("\nType of GroupBy object:")
print(type(grouped))

print("\nGroups based on school code:")

for school, group in grouped:
    print("\nSchool Code:", school)
    print(group)
```

OUTPUT:

192424027

```

IDLE Shell 3.13.9
File Edit Shell Debug Options Window Help
Python 3.13.9 (tags/v3.13.9:8183fa5, Oct 14 2025, 14:09:13) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/hp/AppData/Local/Programs/Python/Python313/16th.py =====
Original DataFrame:
  school class      name date_of_Birth  age  height  weight  address
0  s001     V  Alberto Franco   15/05/2002   12    173     35  street1
1  s002     V   Gino Mcneill   17/05/2002   12    192     32  street2
2  s003    VI   Ryan Parkes   16/02/1999   13    186     33  street3
3  s001    VI   Eesha Hinton   25/09/1998   13    167     30  street1
4  s002     V   Gino Mcneill   11/05/2002   14    151     31  street2
5  s004    VI   David Parkes   15/09/1997   12    159     32  street4

Type of GroupBy object:
<class 'pandas.api.typing.DataFrameGroupBy'>

Groups based on school code:

School Code: s001
  school class      name date_of_Birth  age  height  weight  address
0  s001     V  Alberto Franco   15/05/2002   12    173     35  street1
3  s001    VI   Eesha Hinton   25/09/1998   13    167     30  street1

School Code: s002
  school class      name date_of_Birth  age  height  weight  address
1  s002     V   Gino Mcneill   17/05/2002   12    192     32  street2
4  s002     V   Gino Mcneill   11/05/2002   14    151     31  street2

School Code: s003
  school class      name date_of_Birth  age  height  weight  address
2  s003    VI   Ryan Parkes   16/02/1999   13    186     33  street3

School Code: s004
  school class      name date_of_Birth  age  height  weight  address
5  s004    VI   David Parkes   15/09/1997   12    159     32  street4
>>>

```

17. Write a Pandas program to split the following dataframe by school code and get mean, min, and max value of age for each school.

```

17th.py - C:/Users/hp/AppData/Local/Programs/Python/Python313/17th.py (3.13.9)
File Edit Format Run Options Window Help
import pandas as pd

df = pd.read_csv("C:/Users/hp/Downloads/17TH.csv")

print("Original DataFrame:")
print(df)

result = df.groupby("school")["age"].agg(["mean", "min", "max"])

print("\nMean, Minimum and Maximum Age for each School:")
print(result)

```

```
IDLE Shell 3.13.9
File Edit Shell Debug Options Window Help
Python 3.13.9 (tags/v3.13.9:8183fa5, Oct 14 2025, 14:09:13) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
----- RESTART: C:/Users/hp/AppData/Local/Programs/Python/Python313/17th.py -----
Original DataFrame:
  school class  name date of Birth  age  height  weight  address
0   s001     V  Alberto Franco  15/05/2002   12    173     35  street1
1   s002     V   Gino Mcneill  17/05/2002   12    192     32  street2
2   s003    VI   Ryan Parkes  16/02/1999   13    186     33  street3
3   s001    VI   Eesha Hinton  25/09/1998   13    167     30  street1
4   s002     V   Gino Mcneill  11/05/2002   14    151     31  street2
5   s004    VI   David Parkes  15/09/1997   12    159     32  street4

Mean, Minimum and Maximum Age for each School:
      mean  min  max
school
s001     12.5   12   13
s002     13.0   12   14
s003     13.0   13   13
s004     12.0   12   12
>>>
```

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